🏆 The Definitive MAANG System Design Master Canon
0️⃣ Core Foundations (Non-Negotiable)
Computer Science Fundamentals

- Time & space complexity (incl. amortized analysis)
- Memory hierarchy (CPU cache → RAM → Disk/SSD)
- CPU-bound vs I/O-bound workloads
- Cache locality & data access patterns
  Operating Systems
- Processes vs threads
- Context switching
- Locks, mutexes, semaphores
- Deadlocks & starvation
- Virtual memory basics
  Networking
- TCP vs UDP
- HTTP/1.1 vs HTTP/2 vs HTTP/3
- DNS resolution
- TLS handshake (conceptual)
- Latency vs bandwidth
- Connection pooling
  1️⃣ High-Level Design (HLD)
  1.1 Requirements & Estimation
- Functional vs non-functional requirements
- Latency targets
- Throughput targets
- Availability (e.g., 99.9 vs 99.99)
- Back-of-the-envelope calculations: (QPS, Storage estimation, Memory feasibility, Bandwidth estimation)
  1.2 Data Management & Storage
- Relational Databases: B-Trees (why optimized for disk), Indexing trade-offs, Write-Ahead Logging (WAL), ACID properties, Isolation levels, Read replicas, Master-slave vs multi-writer
- NoSQL Databases: Key-value stores, Document databases, Wide-column stores, When NOT to use NoSQL
- Data Modeling: Normalization vs denormalization, Read-heavy vs write-heavy systems, Hot partitions, Secondary indexes
  1.3 Consistency & Distributed Coordination
- CAP theorem (deep intuition)
- PACELC theorem
- Strong vs eventual consistency
- Quorum (R/W/N)
- Leader election

- Distributed locks
- Saga pattern
- 2-Phase Commit (2PC)
  1.4 Scaling & Partitioning
- Vertical vs horizontal scaling
- Stateless vs stateful services
- Sharding strategies: (Hash-based, Range-based, Geo-based)
- Rebalancing & resharding
- Hot key problem
  1.5 Communication Models & Real-Time Systems ⭐
- Request–Response: REST APIs, gRPC, GraphQL, Idempotency, Pagination, Versioning
- Client-Pull Models: Polling (fixed interval), Long polling
- Server-Push Models: WebSockets (full-duplex), Server-Sent Events (SSE)
- Choosing Between Them: Chat → WebSockets, Live feed → SSE, Periodic dashboard → Polling, Internal service calls → gRPC, Public APIs → REST
  1.6 Caching
- Cache layers (CDN, service, DB)
- LRU, LFU eviction
- Write-through vs write-back
- TTL strategies
- Cache invalidation
- Cache stampede
- Hot keys
  1.7 Asynchronous & Event-Driven Systems
- Message queues vs event streams
- At-most-once / at-least-once / exactly-once
- Idempotent consumers
- Dead-letter queues
- Backpressure handling
- Event-driven architecture
  1.8 Reliability & Fault Tolerance
- Replication
- Redundancy
- Graceful degradation
- Retries with exponential backoff
- Circuit breakers
- Bulkheads
- Cell-based architecture
- Blast radius containment
  1.9 Observability & Operations
- Metrics, logs, distributed tracing

- Liveness vs readiness probes
- SLIs, SLOs, SLAs
- Canary deployments
- Blue-green deployments

1.10 Security

- Authentication vs authorization
- OAuth / JWT (conceptual)
- Rate limiting
- DDoS protection
- Encryption in transit & at rest
- Secrets management

2️⃣ Low-Level Design (LLD)

2.1 Object-Oriented Design

- SOLID principles (applied)
- Encapsulation, abstraction
- Composition vs inheritance
- Immutability
- Cohesion & coupling

2.2 High-Signal Design Patterns

- Creational: Factory, Builder, Singleton (and why dangerous)
- Structural: Adapter, Decorator, Facade, Proxy
- Behavioral: Strategy, Observer, State, Command

2.3 UML & Modeling

- Class diagrams, Sequence diagrams, State diagrams, Entity-relationship modeling

2.4 Concurrency & Thread Safety

- Thread safety principles
- Optimistic vs pessimistic locking
- Atomic operations (CAS intuition)
- Thread pools
- Producer–consumer pattern

2.5 Performance Engineering

- Profiling, Memory leaks, GC basics
- Object lifecycle, Lazy loading, Batching

3️⃣ End-to-End System Design Practice

- URL shortener
- News feed
- Chat system (WebSockets at scale)
- Notification system
- Payment workflow
- File storage system
- For each: APIs, Data model, Caching, Scaling, Failures, Trade-offs

4️⃣ Architect-Level Thinking (The Real Differentiator)

- Latency vs consistency
- Cost vs reliability
- Simplicity vs extensibility
- Strong vs eventual consistency
- Build vs buy decisions
- Cost awareness
- Over-engineering detection
- Failure-first design
- Rollbacks & recovery

5️⃣ Interview Execution Framework

- HLD Flow: Clarify requirements, Define assumptions, Estimate scale, Define APIs, Design data model, High-level architecture, Bottlenecks, Failure handling, Trade-offs
- LLD Flow: Clarify use cases, Identify entities, Design classes, Define relationships, Handle concurrency, Ensure extensibility

🌟 6️⃣ ☐ Elite MAANG Appendices

- 6.1 Advanced Distributed Data & Engines: LSM-Trees vs B-Trees, SSTables & Memtables, Compaction strategies, Online schema changes at scale
- 6.2 Time & Ordering in Distributed Systems: Clock drift, Lamport clocks, Vector clocks, TrueTime
- 6.3 Advanced Networking & Infrastructure: TCP BBR vs Cubic, Anycast & BGP, Edge computing, Service mesh concepts, Sidecar pattern, Control plane vs data plane
- 6.4 Probabilistic & Specialized Data Structures: Bloom filters, HyperLogLog, Quad-trees, Geohashing
- 6.5 High-Scale Data Engineering: Batch vs stream processing, Change Data Capture (CDC), Backpressure & load shedding

🆕 7️⃣ ☐ Staff / Principal-Level Additions

- 7.1 Edge, Gateway & Traffic Management: API gateways, Global load balancing, Request shaping, Throttling, WAF concepts, Edge authentication, Multi-layer rate limiting
- 7.2 Multi-Region & Disaster Recovery: Active-active vs active-passive, RPO & RTO, Cross-region replication lag, Geo-fencing & data residency, Region failover strategies, Split-brain avoidance
- 7.3 Data Lifecycle & Storage Economics: Hot / warm / cold storage tiers, Tiered storage policies, Archival pipelines, Data retention strategies, GDPR-style deletions, Cost-aware storage design
- 7.4 Deployment, Migration & Evolution: Infrastructure as Code mindset, Immutable infrastructure, Feature flags, Dark launches, Schema versioning, Online migrations, Roll-forward vs rollback
- 7.5 Rate Limiting Algorithms (Deep Dive): Token bucket, Leaky bucket, Fixed window, Sliding window log, Sliding window counter, Distributed rate limiting
- 7.6 Advanced Security & Compliance: PII segregation, Encryption key rotation, Audit logging, RBAC vs ABAC, Least privilege enforcement, Zero-trust principles
- 7.7 Advanced Scaling & Migration Patterns: Shadow traffic, Dual writes, Read/write splitting, Strangler Fig pattern, Monolith → microservices evolution, Contract testing
- 7.8 Decision Narration & Staff-Level Communication: Assumption declaration, Multiple-option framing, Explicit trade-off comparison, Justified decision making, Rejected-alternative explanation, Business impact alignment